Machine learning

- Learning is any process by which a system improves performance from experience.

- Machine Learning is a branch of AI that enables computer systems to learns patterns from data and make decisions or predictions.

- Justify the relationship among task - performance → Experience.

⇒ According to tom mitchell, machine Learning is the study of algorithms that improve their performance p at some task t with experience E. The relationship among T, P and E is explained below.

Task: Task is the specific problem or job that the learning system is supposed to perform. It defines what the machine has to do.

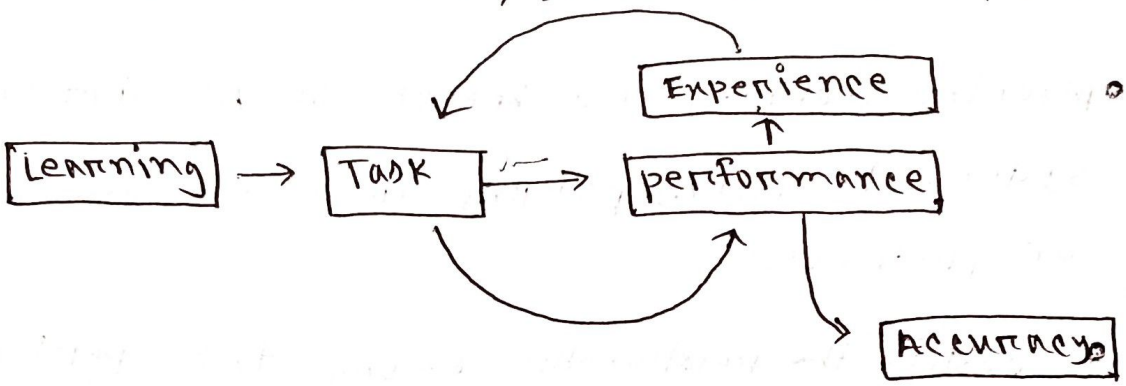
Ex: classifying emails as spam or Legitimate.

performance: performance is the metric used to measure how well the system performs the task.

Ex: percentage of emails correctly classified.

Experience: Experience is the data or training examples from which the system learns.

EX: A database of emails, some with human given labels.

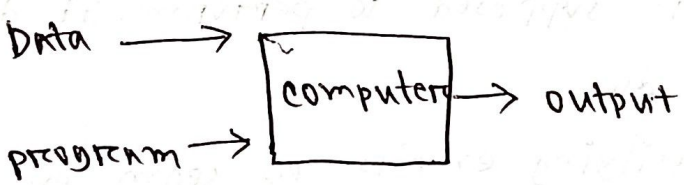


- Distinguish traditional programming and machine Learning.

Traditional programming The computer takes Data and the program as input and produces the output.

In traditional programming, a programmer writes explicit rules by hand. EX: payroll calculation.

Diagram



Machine Learning In machine Learning, we give the computer data and the desired output, and the computer automatically learns the program by finding patterns in the data. EX: spam filter, speech recognition.

Diagram



step of machine learning

1) problem identification
2) Data collection
3) principle component analysis : Feature extraction.
4) pre processing
5) model selection or development
6) Data split (train, test, validation)
7) model evaluation (Accuracy or error rate)
8) model validation (cross validation)
9) Graphical representation.

* Distinguish machine learning steps with example in details.

Feature	supervised	unsupervised	semi supervised	Reinforcement
Datatype	Labeled data	unlabeled data	Few Labeled and many unlabeled	Reward and penalty signal
Goal	predict output from input	Find hidden patterns or groups	Improve learning using both Labeled and unlabeled data	Learn the best action through experience
Output	classification or regression	clusters or patterns	classification or regression	policy (best action)
Human Labels	Required	not required	partially required	Not required
Examples	spam detection, cancer classification	customer segmentation, market analysis	Image classification, with few labeled images	Game playing, Robot navigation

Evaluate the impact of data with the proper understanding and examples.

⇒ Data is the foundation of every machine learning model. The algorithm decides how a model learns, but the data decides what it can learn.

Impact of data on ML models

1) Data quality: More data usually helps a model learn better and make more accurate predictions.

2) Data quality: Clean and correct data is very important. Wrong, missing or duplicate data can reduce model performance.

3) Representativeness and Bias: The dataset should represent all types of users or situations. Biased data can lead to unfair results.

4) Data Drift: If new data becomes different from training data over time, the model's accuracy may decrease.

5) Feature quality: Good features help the model find useful patterns and improve performance.

The success of a machine learning model depends greatly on the quality, quantity and fairness of the data used.

When do we use machine learning ?

⇒ ML is used when

i) Human expertise does not exist (navigating on mars)

ii) Humans can't explain their expertise (speech recognition)

iii) Model must be customized (personalized medicine)

iv) models are based on huge amounts of data (genomics)

* some more examples of tasks that are best solved by using a learning algorithm.

⇒ Recognizing patterns
  ⇒ Facial identities or expression
  ⇒ Medical Images

⇒ Generating patterns
  ⇒ Generating images or motion sequences.

⇒ Recognizing anomalies
  ⇒ Unusual credit card transactions.

⇒ prediction
  ⇒ Future stock prices or currency exchange rates.

Data

Data : Unorganized collection

Information : making organized the data through a process.

methods of data collection :

1) statistical method
2) manual Inspection

Table 1

Name	Colour	Food	Movie	Class
Fariya	white	Burger	My Fault	good
taehyung	red	pizza	Itwarang	good
Jimin	white	Biriyani	Four hands two sonatas	good
Jungkook	Black	Drinks	Mouse	good

Featurename : Name, Colour, Food, movie

Feature values : Fariya, white, burger, myfault etc

predictor feature: A predictor feature is an input variable used to predict the target feature

Target feature : output variables to be predicted.

Data analysis

1) **Data cleaning** : Data cleaning is the process of detecting and correcting errors, missing values, and inconsistencies in data. Data cleaning improves data quality and helps produce accurate machine learning results.

2) **Data preprocessing** The process of converting raw data into mathematical form (Encoding)

3) **statistical measurement :** mean, max, average, quatile, Graphical representation.

4) **Relation analysis :** correlation analysis

5) **principle component analysis :** Identifying the best features, Dimentionality reduction.

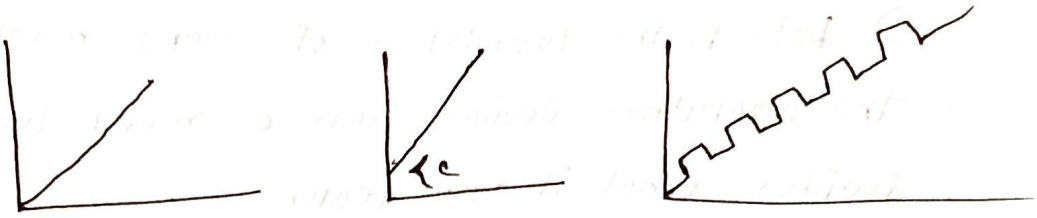
The two main tasks of machine learning are

1. **classification** Dividing input data into proper classes or categories

2. **prediction (Regression) :** predicting an unknown value based on available data.

Linear regression model

$$y = mx + c$$



$$c = \frac{\Sigma y \cdot \Sigma x^2 - \Sigma x \cdot \Sigma xy}{n \cdot \Sigma x^2 - (\Sigma x)^2}$$

$$m = \frac{n \cdot \Sigma xy - \Sigma x \Sigma y}{n \Sigma x^2 - (\Sigma x)^2}$$

Two types of variable

1) Dependent variable
2) Independent variable

→ Question

subject	Age x	Glucose level y
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81

calculate the predicted values and residuals.

$\Rightarrow$

Subject	Age(x)	Glucose level (y)	xy	x^2	y^2
1	43	99	4257	1849	9801
2	21	65	1365	441	4225
3	25	79	1975	625	6241
4	42	75	3150	1764	5625
5	57	87	4959	3249	7569
6	59	81	4779	3481	6561
Σ	247	486	20485	11409	40022

$$C = \frac{\Sigma y \cdot \Sigma x^2 - \Sigma x \cdot \Sigma xy}{n \cdot \Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{486 \cdot 11409 - 247 \cdot 20485}{6 \cdot 11409 - (247)^2}$$

$$= \frac{484979}{7445}$$

$$= 65.14$$

$$m = \frac{n \cdot \Sigma xy - \Sigma x \Sigma y}{n \cdot \Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{6 \cdot 20485 - 247 \cdot 486}{6 \cdot 11409 - (247)^2}$$

$$= \frac{2868}{7445}$$

$$= 0.39$$

x	Actual y	predicted y	Residual
43	99	81.71	17.29
21	65	73.23	-8.23
25	79	74.77	4.23
42	75	81.32	-6.32
57	87	87.10	-0.10
59	81	87.87	-6.87

error calculation

MSE (mean square error)

Formula

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (\text{Residuals})^2$$

MAE (mean Absolute error)

Formula

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |\text{Residuals}|$$

Find-S Algorithm

Hypothesis: A hypothesis is an opinion or statement about a particular subject

2 types of hypothesis :

1) generic hypothesis (parameter – necessary or unnecessary)

11) specific hypothesis : Giving a ~~statement~~ about most important, particular ~~statement~~ parameters

Find-s algorithm The Find-s algorithm works only with positive training examples.

Candidate Elimination Algorithm works with both positive and negative training examples.

Find-s algorithm

1. Initialize h to the most specific hypothesis in H.
2. For each positive training instance x
   For each attribute constraint a_i in h
   If the constraint a_i, is satisfied by x
   Then do nothing
   Else replace a_i, in h by the next more general constraint that is satisfied by x
3. Output hypothesis h

	A	B	C	D	E	F	G
1	sky	airtemp	humidity	wind	water	forecast	enjoy sports
2	sunny	warm	normal	strong	warm	same	yes
3	sunny	warm	high	strong	warm	same	yes
4	rainy	cold	high	strong	warm	change	no
5	sunny	warm	high	strong	cool	change	yes

Ans:

The initial value of hypothesis :

[0, 0, 0, 0, 0, 0]

h1 = [sunny, warm, normal, strong, warm, same]

h2 = [sunny, warm, high ?, strong, warm, same]

h3 = [sunny, warm, high ?, strong, warm, same]

h4 = [sunny, warm, high ?, strong, cool ?, change ?]

If the sky is sunny, airtemp is warm and wind is strong then we enjoy the sports.

specific feature = sky, airtemp, wind

candidate elimination algorithm

- It will provide most generic hypothesis
- positive, negative both samples will be considered
- It is a standard version of Find-S algorithm

Ans

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \rangle$

$S_1 = \langle$ sunny, warm, normal, strong, warm, same $\rangle$

$S_2 = \langle$ sunny, warm, ?, strong, warm, same $\rangle$

$S_3 = \langle$ sunny, warm, ?, strong, warm, same $\rangle$

$S_4 = \langle$ sunny, warm, ?, strong, ?, ? $\rangle$

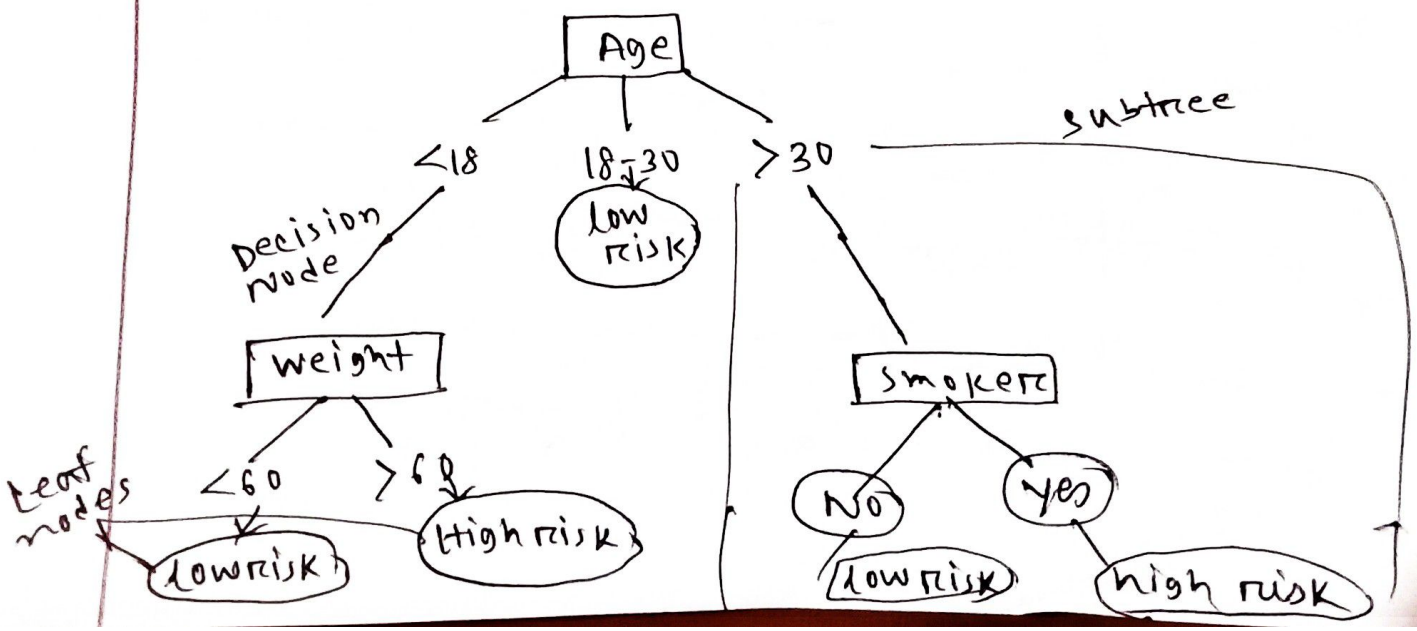
$G_0, G_1, G_2 = \langle ?, ?, ?, ?, ?, ? \rangle$

$G_3, G_4 = \langle$ sunny, ?, ?, ?, ?, ? $\rangle, \langle ?,$ warm, ?, ?, ?, ? $\rangle$

$\langle ?, ?,$ normal, ?, ?, ? $\rangle, \langle ?, ?, ?, ?, ?,$ same $\rangle$

Decision tree algorithm

Different machine learning models have different learning process.



- A Decision tree is a supervised Machine Learning algorithm that uses a tree like structure to make decisions and predictions.
  - * Decision tree can be used for both classification and regression (perform two activities)
  - * A decision tree gives 1 result

- Random forest is a supervised machine learning algorithm that combines multiple decision trees to make prediction.
  - * Random forest is as similar as decision tree. It must have minimum 1000 decision tree.
  - * majority voting happens in random forest.

- How to calculate root node ?
  - => Impurity calculation is used to find the root node.

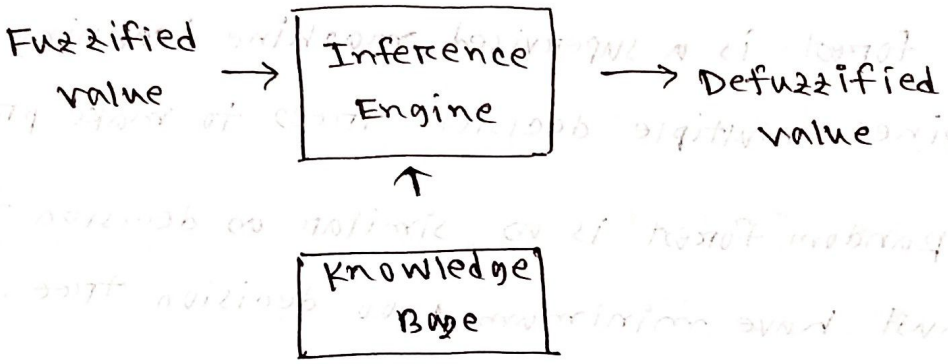
There are two methods for calculating impurity

1) Entropy calculation / Gini Index
2) Information Gain calculation

The feature with the highest information gain is selected as the root node.

Introduction to Expert system

* uniqueness in precision (ES) ES = Expert system
* Fuzzified set / Fuzzified value
* Inference Engine -
* knowledge base.



membership function

$$\mu_A = \square/\square + \square/\square$$

(membership value → input)

$$\mu_A = 0.5/A + 0.6/B + 0.4/C$$

There are 10 types of membership function

A membership function is a special kind of function whose value remains bounded.

$$A = [\mu_A(x)/x] \qquad x \in X$$

Expert system (ES) : It is a computer program or system that acts and solve problems like a human expert is a specific field.

Fuzzified set/ Fuzzified value: Unlike regular computers that only understand strict true(1) or false (0) values, fuzzy logic uses values anywhere between 0 and 1.

Inference Engine: This is the 'brain' of the expert system. It applies logic rules to the information stored in the system to make decisions and find solutions.

knowledge base: This is the library or database where all the expert knowledge, rules and information are stored.

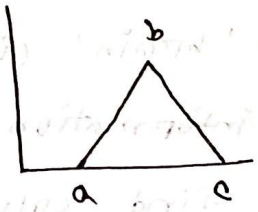
Membership function

1) triangular ~~fun~~ membership function
2) Trapezoid membership function
3) Gaussian " "
4) bellshaped " "
5) sigmoidal " "
6) Left - right " "
7) pi " "
8) open left " "
9) open right " "
10) s - shaped " "

Formula

1) triangular $(x; a, b, c) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \le x < b \\ \frac{c-x}{c-b} & \text{if } b \le x \le c \\ 0 & \text{if } x > c \end{cases}$

shape



a, b, c - constant

2) trapezoid $(x; a, b, c, d) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \le x < b \\ 1 & \text{if } b \le x \le c \\ \frac{d-x}{d-c} & \text{if } c \le x \le d \\ 0 & \text{if } x > d \end{cases}$

3) gaussian $(x; c, \sigma) = e^{-\frac{1}{2}\left(\frac{x-c}{\sigma}\right)^2}$

c = center

σ = standard deviation

4) bell $(x; a, b, c) = \frac{1}{1 + \left|\frac{x-c}{a}\right|^{2b}}$

5) sig $(x; a, c) = \frac{1}{1 + e^{-a(x-c)}}$

6) $LR(x; c, \alpha, \beta) = \begin{cases} F_L\left(\frac{c-x}{\alpha}\right) & \text{if } x \le c \\ F_R\left(\frac{x-c}{\beta}\right) & \text{if } x > c \end{cases}$

7) $pi(x; a, b) = \frac{1}{1+\left(\frac{x-a}{b}\right)^2}$

8) $\text{Open R}(x; \alpha, \beta) = \begin{cases} 0 & \text{if } x < \alpha \\ \frac{x-\alpha}{\beta-\alpha} & \text{if } \alpha \le x \le \beta \\ 1 & \text{if } x > \beta \end{cases}$

9) $\text{open L}(x; \alpha, \beta) = \begin{cases} 1 & \text{if } x < \alpha \\ \frac{\beta-x}{\beta-\alpha} & \text{if } \alpha \le x \le \beta \\ 0 & \text{if } x > \beta \end{cases}$

10) $\text{S-shaped}(x; a, b) = \begin{cases} 0 & \text{if } x < a \\ 2\left(\frac{x-a}{b-a}\right)^2 & \text{if } a \le x < \frac{a+b}{2} \\ 1-2\left(\frac{x-b}{b-a}\right)^2 & \text{if } \frac{a+b}{2} \le x \le b \\ 1 & \text{if } x > b \end{cases}$

- Temperature sensor

$a = 22°C$

$b = 28°C$

$c = 35°C$

$x = \{22, 23, 24, 25, 26, 27, 28, \ldots\ldots 33, 34, 35\}$

Create a fuzzy set for a temperature sensor using a triangular membership function

⇒ Formula :

$$\mu(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \le x < b \\ \frac{c-x}{c-b} & \text{if } b \le x \le c \\ 0 & \text{if } x > c \end{cases}$$

For $22 \le x < 28$

$\mu(x) = (x-22)/(28-22) = (x-22)/6$

$\mu(22) = 0/6 = 0$

$\mu(23) = 1/6 = 0.17$

$\mu(24) = 2/6 = 0.33$

$\mu(25) = 3/6 = 0.50$

$\mu(26) = 4/6 = 0.67$

$\mu(27) = 5/6 = 0.83$

For $28 \leq x \leq 35$

$\mu(x) = (35-x)/(35-28) = (35-x)/7$

$\mu(28) = 7/7 = 1$

$\mu(29) = 6/7 = 0.86$

$\mu(30) = 5/7 = 0.71$

$\mu(31) = 4/7 = 0.57$

$\mu(32) = 3/7 = 0.43$

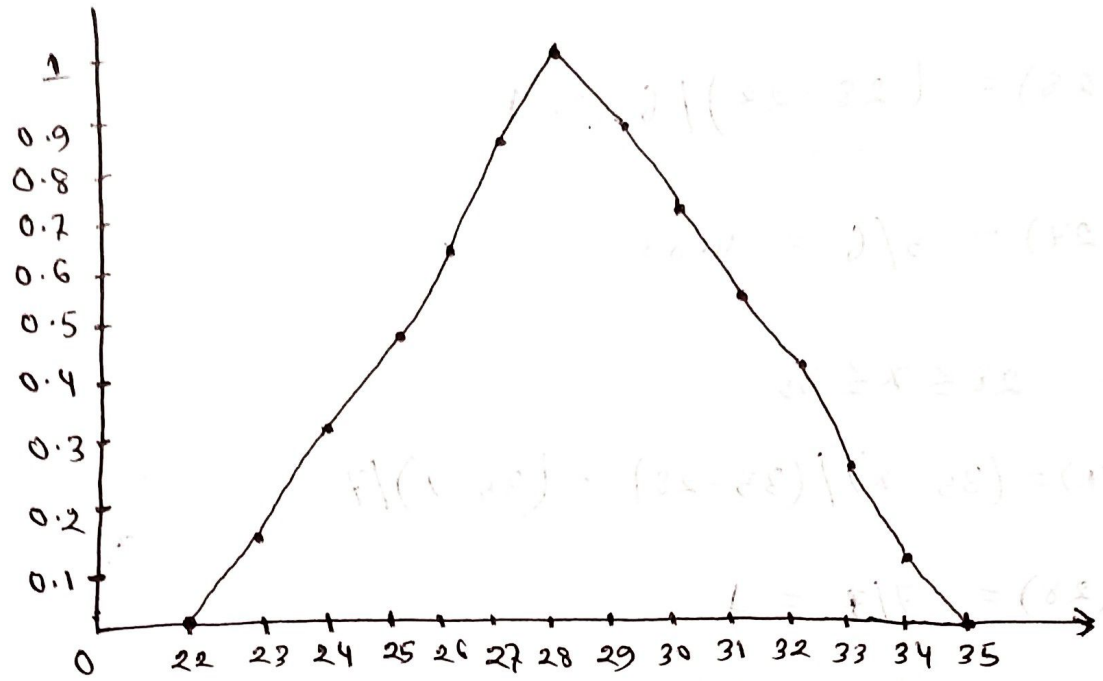
$\mu(33) = 2/7 = 0.29$

$\mu(34) = 1/7 = 0.14$

$\mu(35) = 0/7 = 0$

Fuzzy set, $A = \{ 0/22 + 0.17/23 + 0.33/24 + 0.50/25 + 0.67/26 + 0.83/27 + 1/28 + 0.86/29 + 0.71/30 + 0.57/31 + 0.43/32 + 0.29/33 + 0.14/34 + 0/35 \}$

Graph



3) Gaussian

$$(x; c, \sigma) = e^{-\frac{1}{2}\left(\frac{x-c}{\sigma}\right)^2}$$

Here c = center

σ = standard deviation

$e = 2.71$

EX: $x = \{25, 26, 27, 28, 29, 30\}$

$c = 27.5$

$\sigma = 1.70$

$e = 2.71$

NOW,

$x = 25$, $e = 0.34$ $\left(e^{-\frac{1}{2}\left(\frac{25-27.5}{1.70}\right)^2}\right)$

$x = 26$, $e = 0.67$

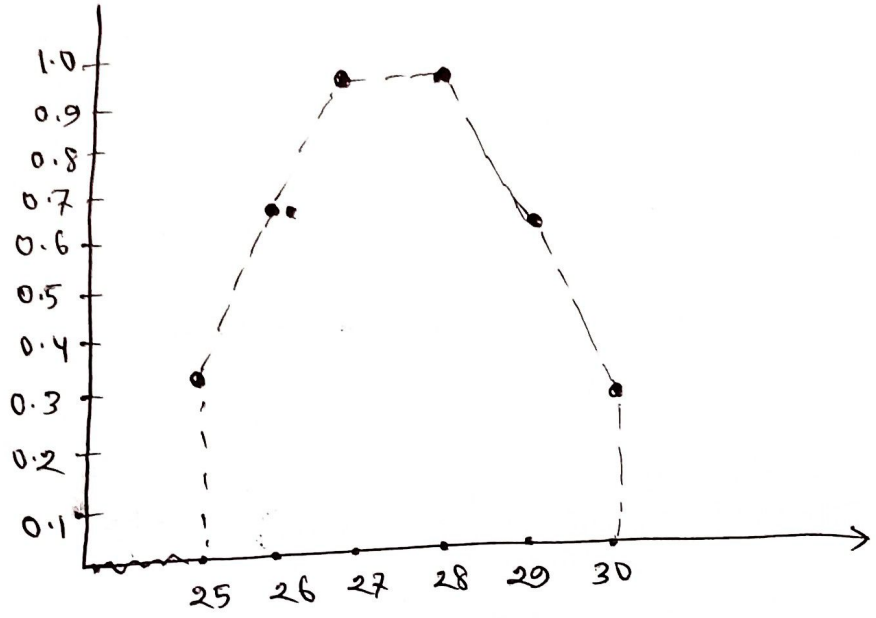
$x = 27$, $e = 0.95$

$x = 28$, $e = 0.95$

$x = 29$, $e = 0.67$

$x = 30$, $e = 0.34$

$$\therefore A = \frac{0.34}{25} + \frac{0.67}{26} + \frac{0.95}{27} + \frac{0.95}{28} + \frac{0.67}{29} + \frac{0.34}{30}$$



Assignment 3

year	Sales (million Euro)	Advertising (million Euro)
1	651	23
2	762	26
3	856	30
4	1063	34
5	1190	43
6	1298	48
7	1421	52
8	1440	57
9	1518	58

Calculate MAE and MSE errors for the given data

Answer

year	Advertising (x)	Sales (y)	xy	x^2	y^2
1	23	651	14973	529	423801
2	26	762	19812	676	580644
3	30	856	25680	900	732736
4	34	1063	36142	1156	1129969
5	43	1190	51170	1849	1416100
6	48	1298	62304	2304	1684804
7	52	1421	73892	2704	2019241
8	57	1440	82080	3249	2073600
9	58	1518	88044	3364	2304324
Σ	371	10199	454097	16731	12365219

$$M = \frac{n\Sigma xy - \Sigma x \Sigma y}{n \cdot \Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{9 \times 454097 - 371 \times 10199}{9 \times 16731 - (371)^2}$$

$$= 23.42$$

$$C = \frac{\Sigma y \cdot \Sigma x^2 - \Sigma x \cdot \Sigma xy}{n \cdot \Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{10199 \times 16731 - 371 \times 454097}{9 \times 16731 - (371)^2}$$

$= 167.68$

x	Actual, y	predicted, y	Residuals	(Residuals)^2
23	651	706.34	-55.34	3062.52
26	762	776.6	-14.6	213.16
30	856	870.28	-14.28	203.92
34	1063	963.96	99.04	9808.92
43	1190	1174.74	15.26	232.87
48	1298	1291.84	6.16	37.95
52	1421	1385.52	35.48	1258.83
57	1440	1502.62	-62.62	3921.26
58	1518	1526.04	-8.04	64.64

$\therefore \Sigma |\text{Residuals}| = 310.82$

$\therefore \Sigma (\text{Residuals})^2 = 18804.07$

$\therefore \text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |\text{Residuals}| = \frac{1}{9} \times 310.82 = 34.524$

$\therefore \text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (\text{Residuals})^2 = \frac{1}{9} \times 18804.07 = 2089.34$

(Ans)

play Tennis : Training examples

1)

Day	outlook	Temperature	Humidity	Wind	playtennis
D1	sunny	Hot	High	weak	NO
D2	sunny	Hot	High	strong	NO
D3	overcast	Hot	High	weak	yes
D4	Rain	mild	High	weak	yes
D5	Rain	cool	Normal	weak	yes
D6	Rain	cool	Normal	strong	NO
D7	overcast	cool	Normal	strong	yes
D8	sunny	mild	High	weak	NO
D9	sunny	cool	Normal	weak	yes
D10	Rain	mild	Normal	weak	yes
D11	sunny	mild	Normal	strong	yes
D12	overcast	mild	High	strong	yes
D13	overcast	Hot	Normal	weak	yes
D14	Rain	mild	High	strong	NO

construct a decision tree for the 'play tennis' dataset using information Gain. Identify the root node and show all calculations.

⇒ Total sample = 14

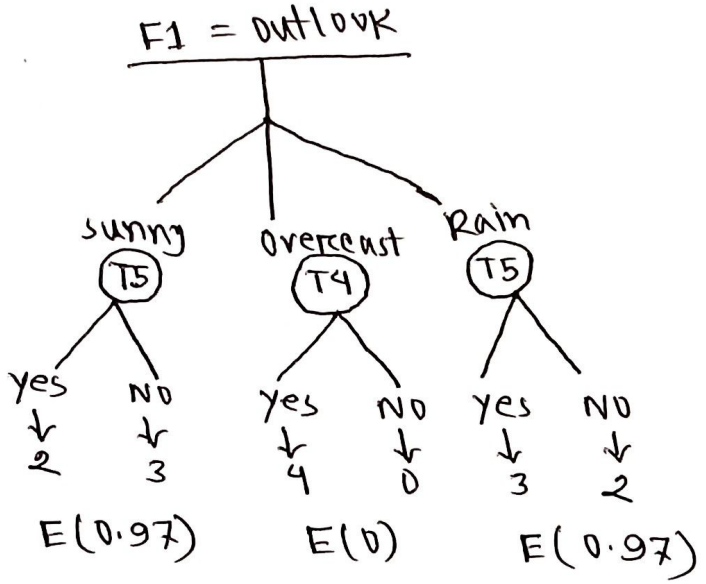
yes = 9

NO = 5

$$\text{Entropy} = \sum_{i=0} -P_i \log_2 P_i$$

$$= -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14}$$

$$= 0.94$$



for sunny,

$$\text{Entropy} = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5}$$

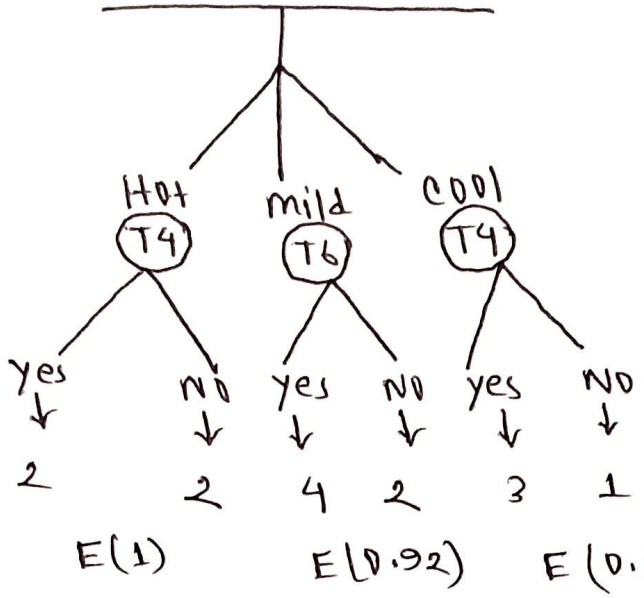
$$= 0.97$$

Information gain, for outlook,

$$= 0.94 - \left[\frac{5}{14} \times 0.97 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.97\right]$$

$$= 0.25$$

F2- Temperature



For mild,

$$\text{Entropy} = -\frac{4}{6}\log_2\frac{4}{6} - \frac{2}{6}\log_2\frac{2}{6}$$

$$= 0.92$$

for cool,

$$\text{Entropy} = -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4}$$

$$= 0.81$$

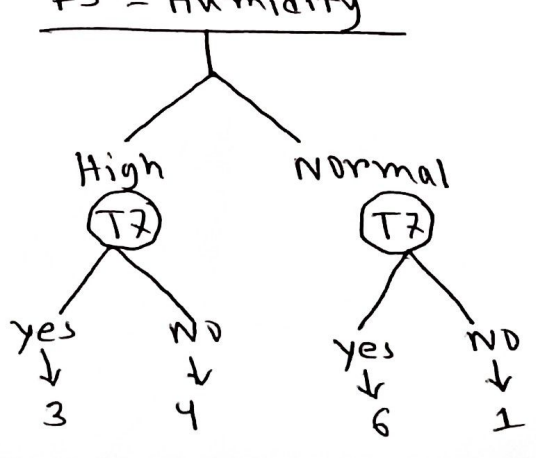
Information gain,

for Temperature,

$$= 0.94 - \left[\frac{4}{14} \times 1 + \frac{6}{14} \times 0.92 + \frac{4}{14} \times 0.81\right]$$

$$= 0.03$$

F3 - Humidity



For High,

$$\text{Entropy} = -\frac{3}{7}\log_2\frac{3}{7} - \frac{4}{7}\log_2\frac{4}{7}$$

$$= 0.99$$

For normal,

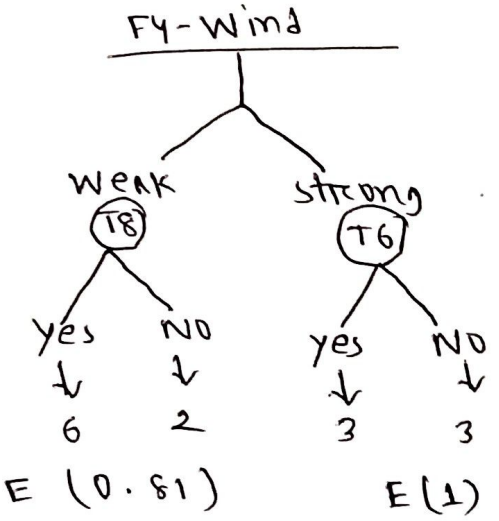
$$\text{Entropy} = -\frac{6}{7}\log_2\frac{6}{7} - \frac{1}{7}\log_2\frac{1}{7}$$

$$= 0.59$$

Information gain, for Humidity,

$$= 0.94 - \left[\frac{7}{14} \times 0.99 + \frac{7}{14} \times 0.59\right)$$

$$= 0.15$$



For weak,

$$\text{Entropy} = -\frac{6}{8}\log_2\frac{6}{8} - \frac{2}{8}\log_2\frac{2}{8}$$

$$= 0.81$$

Information gain, for Wind,

$$= 0.94 - \left[\frac{8}{14} \times 0.81 - \frac{6}{14} \times 1\right]$$

$$= 0.05$$

∴ Outlook has the highest Information Gain, so Outlook is the root node.

Splitting the Branches :

Overcast branch : all 4 samples are yes, so it becomes the leaf yes.

For sunny branch

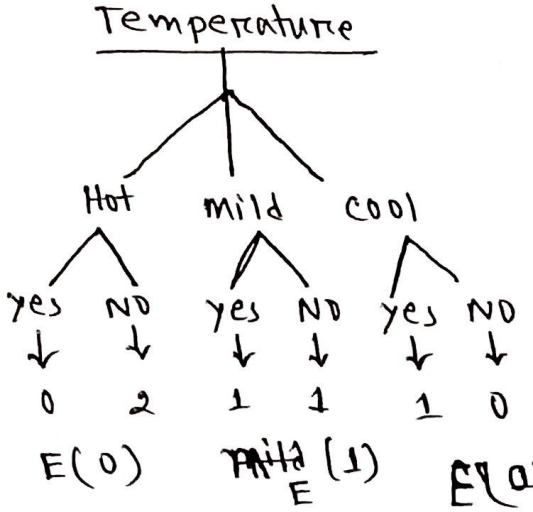
Outlook	Temperature	Humidity	Wind	playtennis
sunny	Hot	High	weak	NO
sunny	Hot	High	strong	NO
sunny	mild	High	weak	NO
sunny	cool	normal	weak	yes
sunny	mild	Normal	strong	yes

total sample = 5

yes = 2

NO = 3

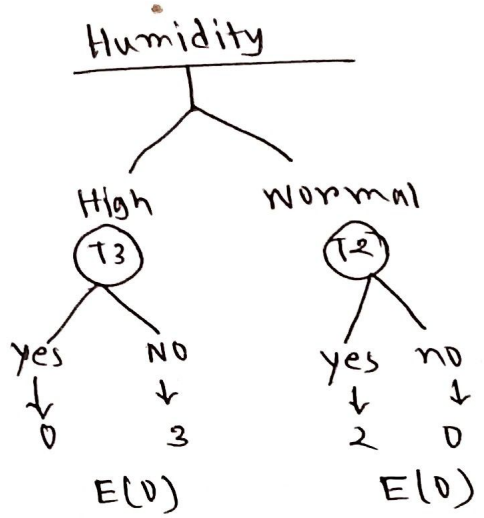
Entropy (0.97)



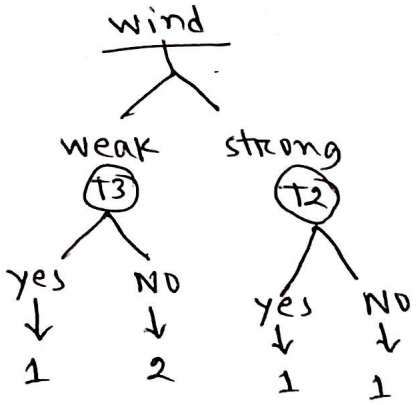
Information Gain,

$$= 0.97 - \left[\frac{2}{5} \times 0 + \frac{2}{5} \times 1 + \frac{1}{5} \times 0\right]$$

$$= 0.57$$



Information Gain = 0.97



For weak,

$$\text{Entropy} = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}$$

$$= 0.92$$

Information Gain, for wind,

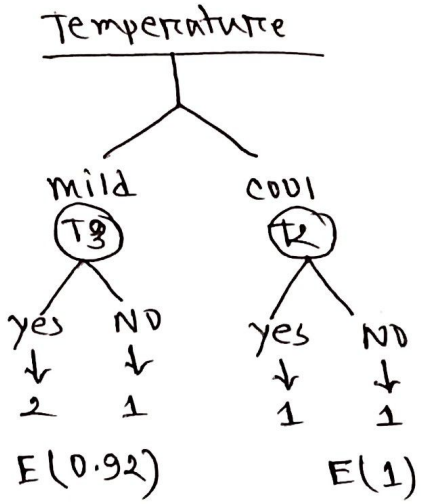
$$= 0.97 - \left(\frac{3}{5} \times 0.92 + \frac{2}{5} \times 1\right)$$

$$= 0.02$$

∴ Humidity is selected, and both of its branches are pure.

Rain branch

Outlook	Temperature	Humidity	Wind	playtennis
Rain	mild	High	weak	yes
Rain	cool	Normal	weak	yes
Rain	cool	Normal	strong	NO
Rain	mild	Normal	weak	yes
Rain	mild	High	strong	NO



. Total sample = 5
yes = 3
NO = 2
E ≠ 0.97)
for Rain branch

For mild,

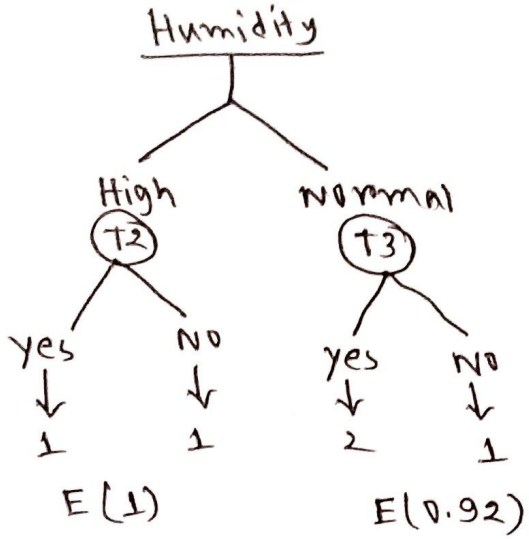
$$\text{Entropy} = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$$= 0.92$$

Information Gain, for temperature

$$= 0.97 - \left(\frac{3}{5} \times 0.92 + \frac{2}{5} \times 1\right)$$

$$= 0.02$$



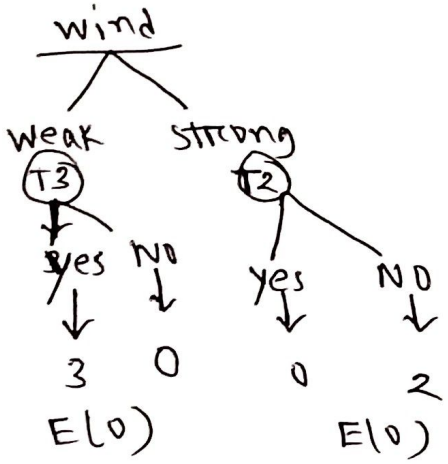
For Normal,

$$\text{Entropy} = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$$= 0.92$$

Information Gain, for Humidity

$$= 0.02$$



Information Gain = 0.97

∴ wind is selected, and both of its branches are pure.

Final Decision Tree

